

Proof: By defn, if there exists real weights $c_1, c_2, \dots, c_p$ where all are not zero together, and $c_1 v_1 + c_2 v_2 + \dots + c_p v_p = 0$.

Let, $\exists i$ where $1 \leq i \leq p$ s.t. $v_i = 0$. $\rightarrow 0$

$$\therefore c_1 v_1 + c_2 v_2 + \dots + c_{i-1} v_{i-1} + \boxed{c_i v_i} + c_{i+1} v_{i+1} + \dots + c_p v_p = \boxed{0}$$

From Theorem 7, the vectors $\{v_1, v_2, \dots, v_p\}$ are linearly dependent. $\square$

Example 6: Determine by inspection if the given set is linearly independent.

a) $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

Soln: No. of vectors = 4 / No. of entries in each vector = 3

By Theorem 8, the given set is linearly dependent.

b) $\begin{bmatrix} 2 \\ 9 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$

Soln: A zero vector is present. By Theorem 9, the given set is linearly dependent.

c) $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ -9 \\ 15 \end{bmatrix}$

Soln: Compare the corresponding entries of the two vectors. The second vector seems to be $-3/2$ times the first vector. This relation holds for the first three pairs of entries, but fails for the fourth pair. Thus neither of the vectors is a multiple of the other, and hence they are linearly independent. $\square$

PROOF OF THEOREM 7 (Characterization of Linearly Dependent Sets)

Let an indexed set $S = \{v_1, v_2, \dots, v_p\}$ be linearly dependent. By defn, $\exists$ weights $c_1, c_2, \dots, c_p$ such that $\boxed{c_1 v_1 + c_2 v_2 + \dots + c_p v_p = 0}$, where $c_1, c_2, \dots, c_p$ are not all 0 together.